

Woongjib Choi

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RESEARCH INTERESTS

Digital Signal Processing, Speech Signal Processing, Audio Compression, Audio Generation, Generative Models

EDUCATION

Yonsei University Mar. 2024 – Feb. 2026 (Expected)
M.S. in Electrical and Electronic Engineering (Advisor: Prof. Hong-Goo Kang) Seoul, South Korea

Yonsei University Mar. 2018 – Feb. 2024
B.S. in Electrical and Electronic Engineering Seoul, South Korea
• GPA: 4.0 / 4.5

EXPERIENCE

Researcher (Project Leader, Final 6 Months) Mar. 2024 – Dec. 2025
ETRI-Yonsei Joint Research Project Seoul, South Korea

- Collaborated with Electronics and Telecommunications Research Institute (ETRI) to develop DNN-based post-processing modules for legacy audio codecs.
- Led project coordination in final 6 months, managing joint research presentations, technical reports, and facilitating inter-institutional communications.
- Research resulted in 1 international conference paper, 1 domestic conference paper, and 1 patent application.

Teaching Assistant Mar. 2024 – Jun. 2025
Yonsei University, DSPAI Lab Seoul, South Korea

- **EEE2111: Analog Electronics Lab** (2025-1)
- **EEE2060: Signals and Systems** (2024-2)
- **EEE4610: Electrical and Electronic Engineering Capstone Design** (2024-2)
 - * Advised undergraduate capstone project: *Unified Speech Enhancement: Bridging Personalized and Unconditional Model*
- **EEE4420: Digital Signal Processing** (2024-1)

Undergraduate Research Intern Jun. 2023 – Jan. 2024
Yonsei University, SRIP Lab (Supervisor: Prof. Moon Gi Kang) Seoul, South Korea

- Implemented and analyzed image processing algorithms: white balancing, denoising, color interpolation (demosaicing), and super-resolution.

PUBLICATIONS

International Conferences

- [1] **Woongjib Choi**, Sangmin Lee, Hyungseob Lim, and Hong-Goo Kang, "UniverSR: Unified and Versatile Audio Super-Resolution via Vocoder-Free Flow Matching." *Submitted to ICASSP 2026*.
- [2] Jaehun Shin, **Woongjib Choi**, Jihyun Lee, Hyungseob Lim, Inseon Jang, and Hong-Goo Kang, "Neural Joint Stereo Coding for High Spatial Fidelity via Explicit Binaural Cue Loss Functions." *Submitted to ICASSP 2026*.
- [3] Sangmin Lee, **Woongjib Choi**, Jihyun Kim, and Hong-Goo Kang, "SAGE-LD: Towards Scalable and Generalizable End-to-End Language Diarization via Simulated Data Augmentation." *Submitted to ICASSP 2026*.
- [4] **Woongjib Choi**, Byeong Hyeon Kim, Hyungseob Lim, Inseon Jang, and Hong-Goo Kang, "Neural Spectral Band Generation for Audio Coding." in *Proceedings of INTERSPEECH*, 2025.

Domestic Conferences

- [1] Jaehun Shin, **Woongjib Choi**, Byeong Hyeon Kim, Inseon Jang, and Hong-Goo Kang, "DNN-based Speech Codec Enhancement Utilizing Conditional Flow Matching." in *Proceedings of the Korean Institute of Broadcast and Media Engineers (KIBME) Conference*, 2025.
- [2] **Woongjib Choi**, Byeong Hyeon Kim, and Hong-Goo Kang, "Non-Blind Speech Bandwidth Extension Using Pre-Trained Self-Supervised Features." in *Proceedings of the Institute of Electronics and Information Engineers (IEIE) Conference*, 2024.

PROJECTS

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| Context-Aware Time-Scale Modification (TSM) with LLM Guidance | Dec. 2024 |
| <ul style="list-style-type: none">• Final project for the <i>Speech Signal Processing</i> course (EEE5340).• Developed a TSM system that combines the traditional SOLA algorithm with an LLM to selectively control speech speed in important frames. | |
| LAUD: Large Language Model as a Universal ASR Decoder | Jun. 2024 |
| <ul style="list-style-type: none">• Final project for the <i>Speech Interface</i> course (EEE7341).• Developed a multilingual ASR framework that utilizes a speech foundation model with the Roman alphabet as a universal intermediate representation. | |
| A Technical Overview of Deep Learning Based Single-Image Super-Resolution Methods | Dec. 2023 |
| <ul style="list-style-type: none">• Undergraduate Capstone Project (EEE4610).• Investigated and compared three major categories on single-image super-resolution: external learning-based (e.g., SRGAN), internal learning-based (e.g., ZSSR), and Transformer-based approaches (e.g., SwinIR). | |

AWARDS & HONORS

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| Dean's List | Spring 2023 |
| <i>Yonsei University</i> | Seoul, South Korea |
| <ul style="list-style-type: none">• Awarded for high academic achievement, ranking within the top 3% of the college. | |
| Academic Excellence Scholarship | 2023 |
| <i>Yonsei University</i> | Seoul, South Korea |
| <ul style="list-style-type: none">• Recipient for both the Spring 2023 and Fall 2023 semesters based on academic excellence. | |

SKILLS

- Programming Languages:** Python, MATLAB, C/C++, Verilog
- Frameworks & Libraries:** PyTorch, torchaudio, NumPy, SciPy, Librosa
- Tools:** Linux Shell, Git, Arduino, Xilinx Vitis/Vivado

EXTRACURRICULAR ACTIVITIES

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|---|-----------------------|
| Editor-in-Chief | Mar. 2022 – Jun. 2023 |
| <i>Humans of Yonsei (Student Storytelling and Interview Organization)</i> | Seoul, South Korea |
| Sergeant, Republic of Korea Army | Apr. 2020 – Oct. 2021 |
| <i>Repair Parts and Tool Supply Specialist (Mandatory Military Service)</i> | Pocheon, South Korea |
| Freshman Class Representative | Mar. 2018 – Sep. 2018 |
| <i>Department of Electrical & Electronic Engineering, Yonsei University</i> | Seoul, South Korea |